

Project Progress - Week 1 of March 2025

March 4, 2025

This is a file that contains the tracking of the activities related to the OFC project during the first week of March 2025. The activities are divided into groups and a summary of the progress is provided.

1 Initial Status of the Project

The project has made significant progress up until the first week of March 2025. The key achievements include:

- Successful implementation and validation of the optical force calculation function in the dipole approximation.
- Development of test and validation scripts that confirm the correct dependence on the permittivity.
- Advancement in the theoretical documentation, particularly in the section on polarizability.
- Detailed derivation of the quasi-static polarizability formula for a dielectric sphere.

2 Monday

The repository structure has been fixed and the management of Python modules has been addressed. Additionally, the team has been learning how to compile LaTeX projects located inside the repository. This document serves as an example of the compilation process.